

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A right circular cylinder has a base diameter of **22** centimeters and a height of **6** centimeters. What is the volume, in cubic centimeters, of the cylinder?

Answer

- A. **132** π
- B. **264** π
- C. **726** π
- D. **2,904** π

Correct Answer: C

Rationale

Choice C is correct. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. It's given that a right circular cylinder has a height of **6** centimeters. Therefore, $h = 6$. It's also given that the cylinder has a base diameter of **22** centimeters. The radius of a circle is half the diameter of the circle. Since the base of a right circular cylinder is a circle, it follows that the radius of the base of the right circular cylinder is $\frac{22}{2}$, or **11**, centimeters. Therefore, $r = 11$. Substituting **11** for r and **6** for h in the formula $V = \pi r^2 h$ yields $V = \pi(11)^2(6)$, which is equivalent to $V = \pi(121)(6)$, or $V = 726\pi$. Therefore, the volume, in cubic centimeters, of the cylinder is **726** π .

Choice A is incorrect. This is the volume of a right circular cylinder that has a base diameter of $2\sqrt{22}$, not **22**, centimeters and a height of **6** centimeters.

Choice B is incorrect. This is the volume of a right circular cylinder that has a base diameter of $4\sqrt{11}$, not **22**, centimeters and a height of **6** centimeters.

Choice D is incorrect. This is the volume of a right circular cylinder that has a base diameter of **44**, not **22**, centimeters and a height of **6** centimeters.